

* Scaled Dot Product Attention

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_K}}\right) \cdot V$$

• What is $\frac{1}{\sqrt{d_K}}$?

→ Scaling factor, which solves unstable gradient problem

→ d_K : dimension of Key vector

Key vector = $[0.1 \quad 0.2 \quad 0.7 \quad 0.9]$

$\therefore d_K = \text{dim of } K = 4$ $\therefore d_K = 4$

• What was the need to divide it by $\frac{1}{\sqrt{d_K}}$?

→ To solve the problem of high variance!!

• If dim is low $\rightarrow$ variance is low

• If dim is high $\rightarrow$ variance is high

Eg $K = [10 \quad 20 \quad 30 \quad 40 \quad 50 \quad 60 \quad 70]$

$\therefore \text{variance} = 400$

~~to solve~~

To lower this variance, we need to scale down the numbers

$K = [1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7]$

$\therefore \text{variance} = 4$

$\therefore$ To lower ^{the} variance we need to scale it down the numbers without reducing the ~~#~~ dimension

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What is the problem when variance is high?

- If variance is high numbers are distributed such that their absolute difference is high.
- So during back propagation, the numerical value which has high ^{value}, the algorithm would try to minimize it only i.e. the algorithm's focus would shift in optimizing higher value number, thereby neglecting smaller value number which leads to poor training.
- Hence variance should be low.

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So, how is a particular value is used for

Scaling?

Eg $x = \begin{bmatrix} a \\ b \end{bmatrix} = a \cdot b \rightarrow \therefore \text{var}(x) = \text{var}(x)$

2) $y = \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} c & d \\ e & f \end{bmatrix} = \begin{bmatrix} ac+bd \\ ae+bf \end{bmatrix} \rightarrow \text{var}(y) \approx 2 \times \text{var}(x)$

3) $z = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} d & e & f \\ g & h & i \end{bmatrix} \rightarrow \begin{bmatrix} ad+be+cf \\ ag+bh+ci \end{bmatrix} \rightarrow \therefore \text{var}(z) \approx 3 \times \text{var}(x)$

So, it's a linear relation if $\text{dim} \uparrow$, variance $\uparrow$
If you multiply a variable X is constant c to form a variable Y , such that $Y = cX$, then
 $\text{var}(Y) = c^2 \text{var}(X)$

$$\therefore \text{Eg: } X \rightarrow \text{var}(x) \\ cX \rightarrow c^2 \text{var}(x)$$

Now we
are
getting
same
variance

$$Y \rightarrow \text{var}(Y) \rightarrow \approx 2 \times \text{var}(x)$$

$$\frac{Y}{\sqrt{2}} \rightarrow \frac{1}{2} \times 2 \times \text{var}(x) \rightarrow \text{var}(x)$$

$$\frac{Z}{\sqrt{3}} \rightarrow \frac{1}{3} \times 3 \times \text{var}(x) \rightarrow \text{var}(x)$$

So, by scaling it down to $\frac{1}{\sqrt{d_K}}$, we are setting high dimensional vector's variance equal to smaller (1 or 2nd) dimensional vector, for ~~get~~ better training

* Difference b/w Self Attention & Luong Attention

- The procedure of calculating vector for both self and Luong attention is same, so self and Luong attention are same only.
- In Luong attention, attention is ~~cal~~ calculated b/w self and inter (another) sequence.
- In Self attention, attention is calculated b/w the sequence of it self.

$$S_1 + S_2$$

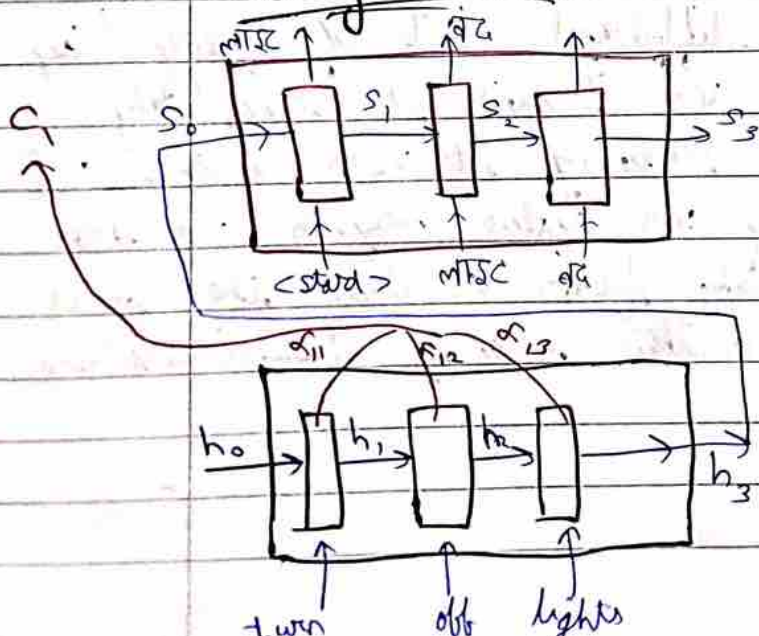
Attention ~~cal~~ calculation = Luong attention

$$S_1 + S_1 (\text{Itself})$$

Attention calculation = self attention

∴ That is why it is called self attention.

Luong attention



$$C_1 = \alpha_{11} h_1 + \alpha_{12} h_2 + \alpha_{13} h_3$$

$$C_2 = \alpha_{21} h_1 + \alpha_{22} h_2 + \alpha_{23} h_3$$

$$\alpha_{ij} = \text{Softmax}(e_{ij})$$

$$e_{ij} = S_i^T h_j$$

here: h_j : Query vector
 S_i^T : Key vector
 $K^T Q \approx Q \cdot K^T$